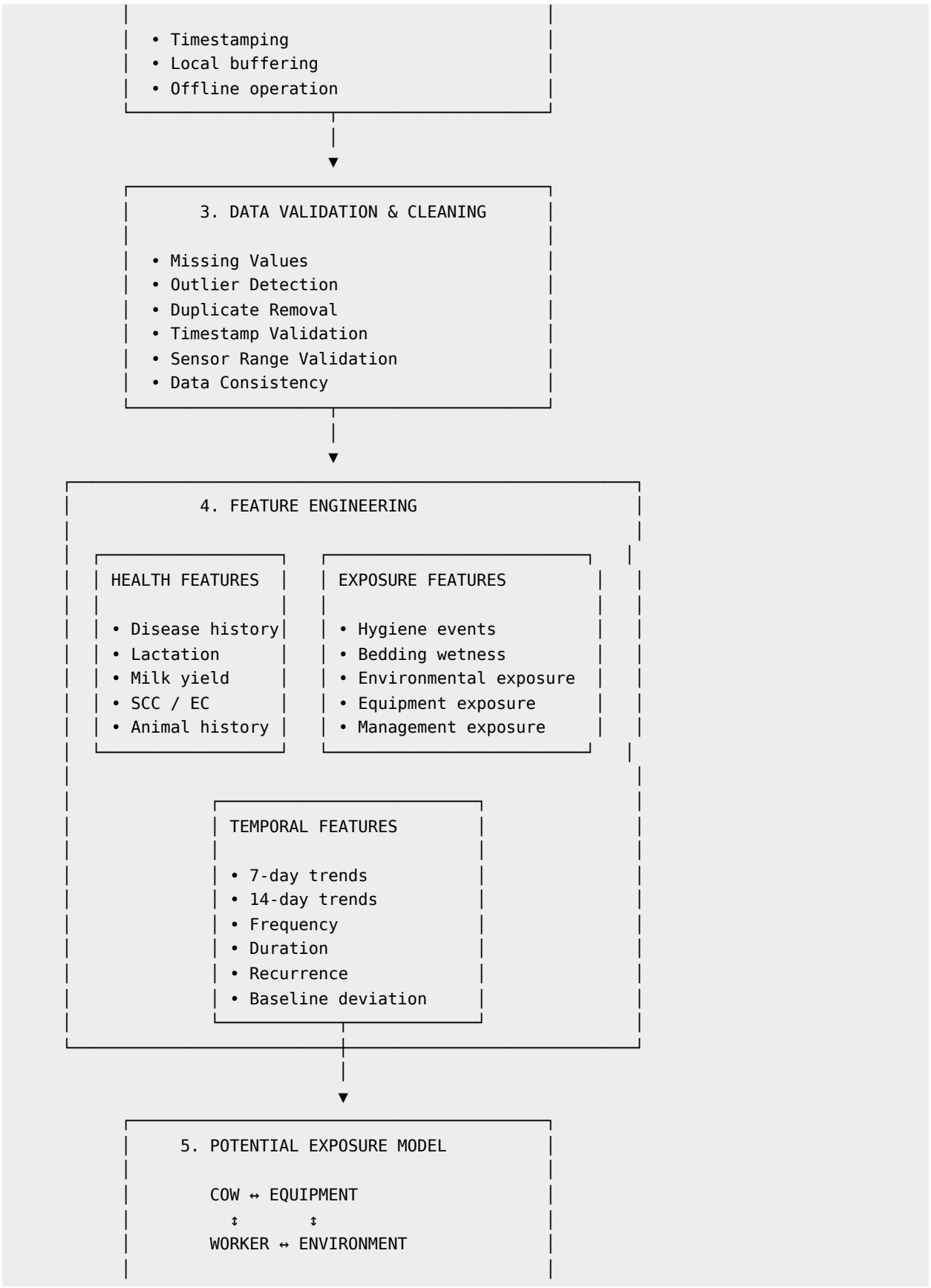
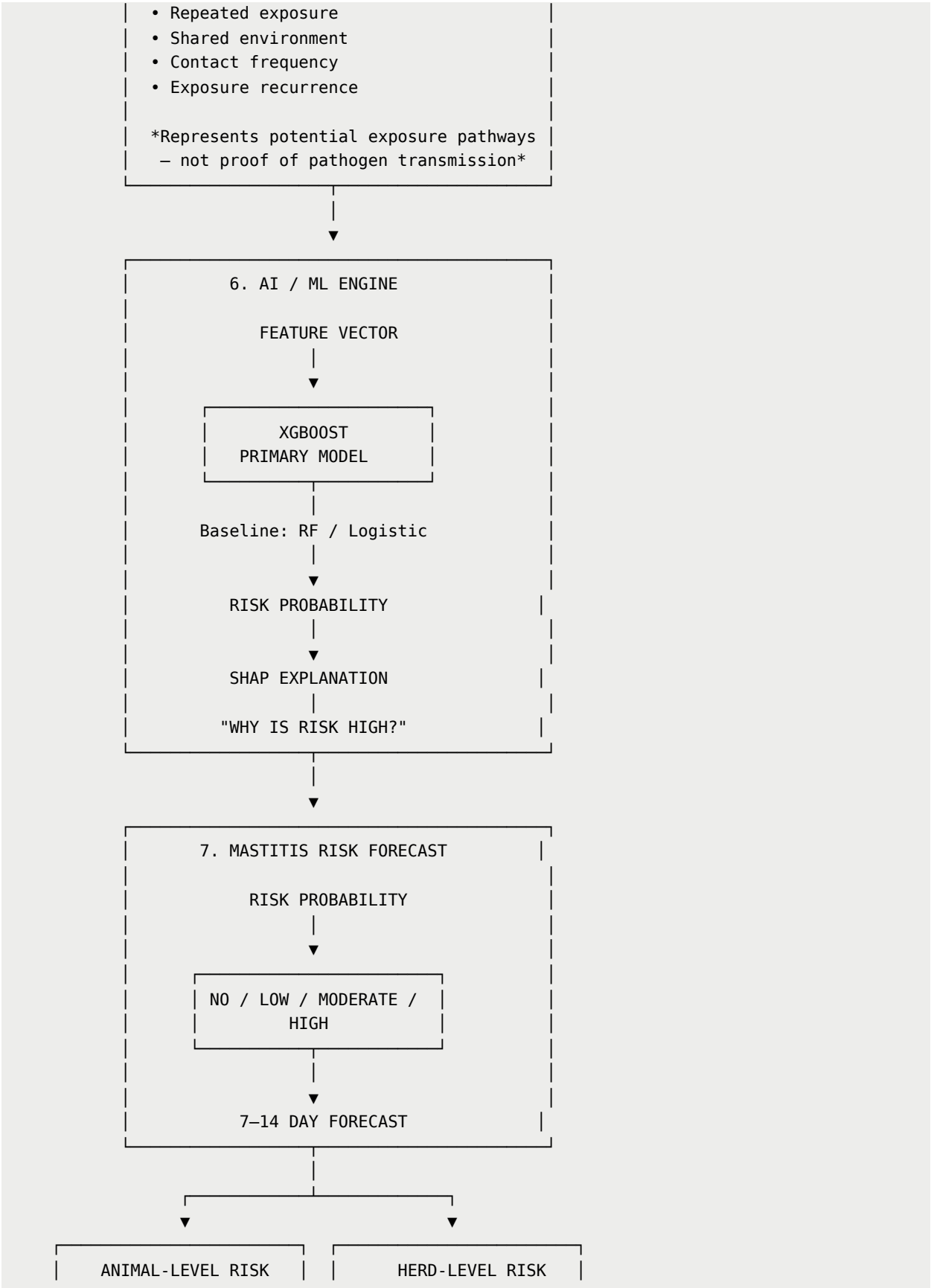
Yes. Based on the roast, I would make the final architecture much cleaner and keep the core story focused on:

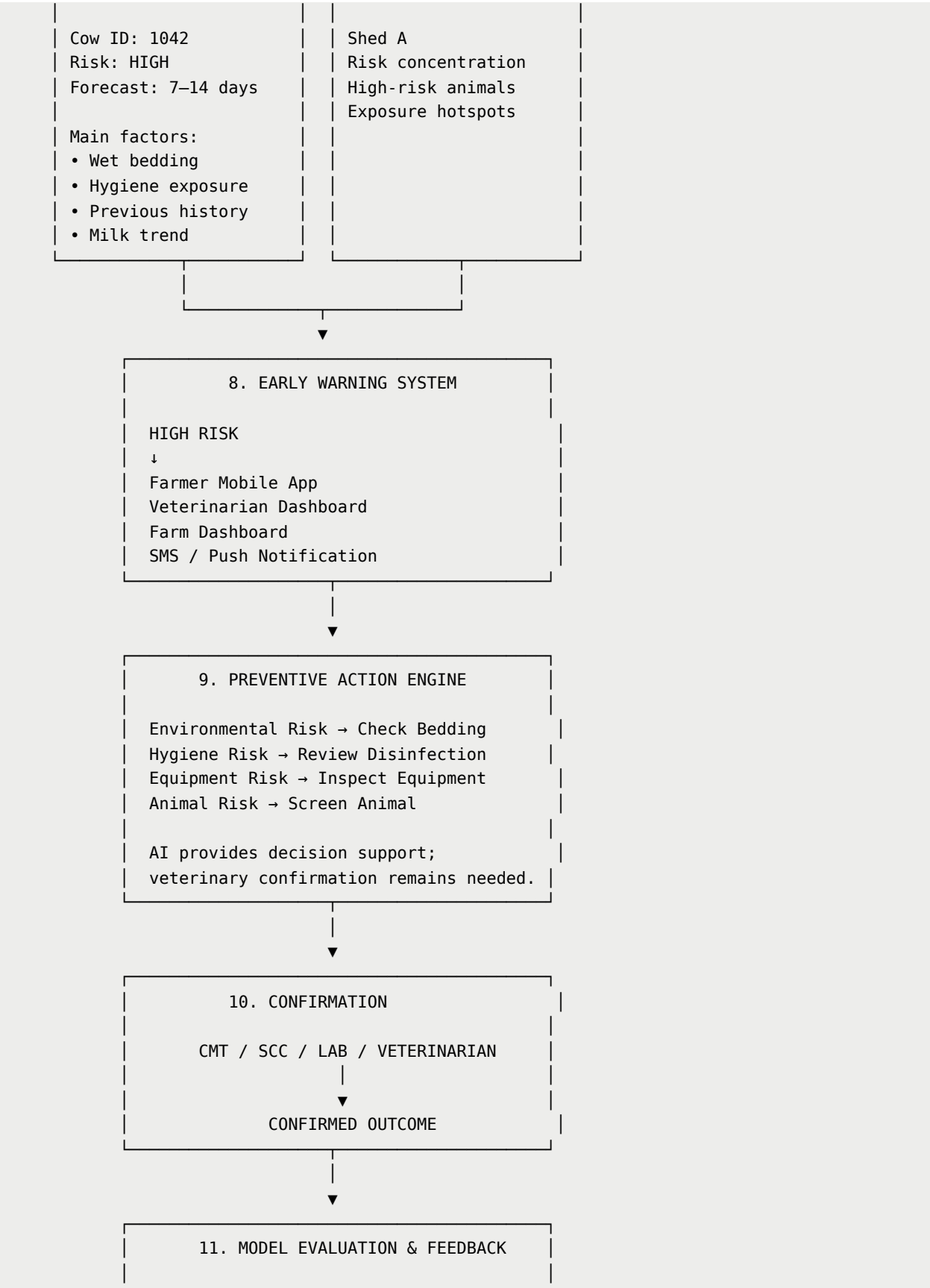
Collect → Validate → Build Exposure/Health Features → Temporal ML → Predict → Alert → Confirm → Learn

FINAL FLOWCHART — SIH PS 26109




• Timestamping
• Local buffering
• Offline operation
3. DATA VALIDATION & CLEANING
• Missing Values
• Outlier Detection
• Duplicate Removal
• Timestamp Validation
• Sensor Range Validation
• Data Consistency
4. FEATURE ENGINEERING
HEALTH FEATURES
• Disease history
• Lactation
• Milk yield
• SCC / EC
• Animal history
EXPOSURE FEATURES
• Hygiene events
• Bedding wetness
• Environmental exposure
• Equipment exposure
• Management exposure
TEMPORAL FEATURES
• 7-day trends
• 14-day trends
• Frequency
• Duration
• Recurrence
• Baseline deviation
5. POTENTIAL EXPOSURE MODEL
COW ↔ EQUIPMENT
↕ ↕
WORKER ↔ ENVIRONMENT


• Repeated exposure
• Shared environment
• Contact frequency
• Exposure recurrence
Represents potential exposure pathways – not proof of pathogen transmission
6. AI / ML ENGINE
FEATURE VECTOR
XGBOOST
PRIMARY MODEL
Baseline: RF / Logistic
RISK PROBABILITY
SHAP EXPLANATION
"WHY IS RISK HIGH?"
7. MASTITIS RISK FORECAST
RISK PROBABILITY
NO / LOW / MODERATE / HIGH
7–14 DAY FORECAST
ANIMAL-LEVEL RISK
HERD-LEVEL RISK

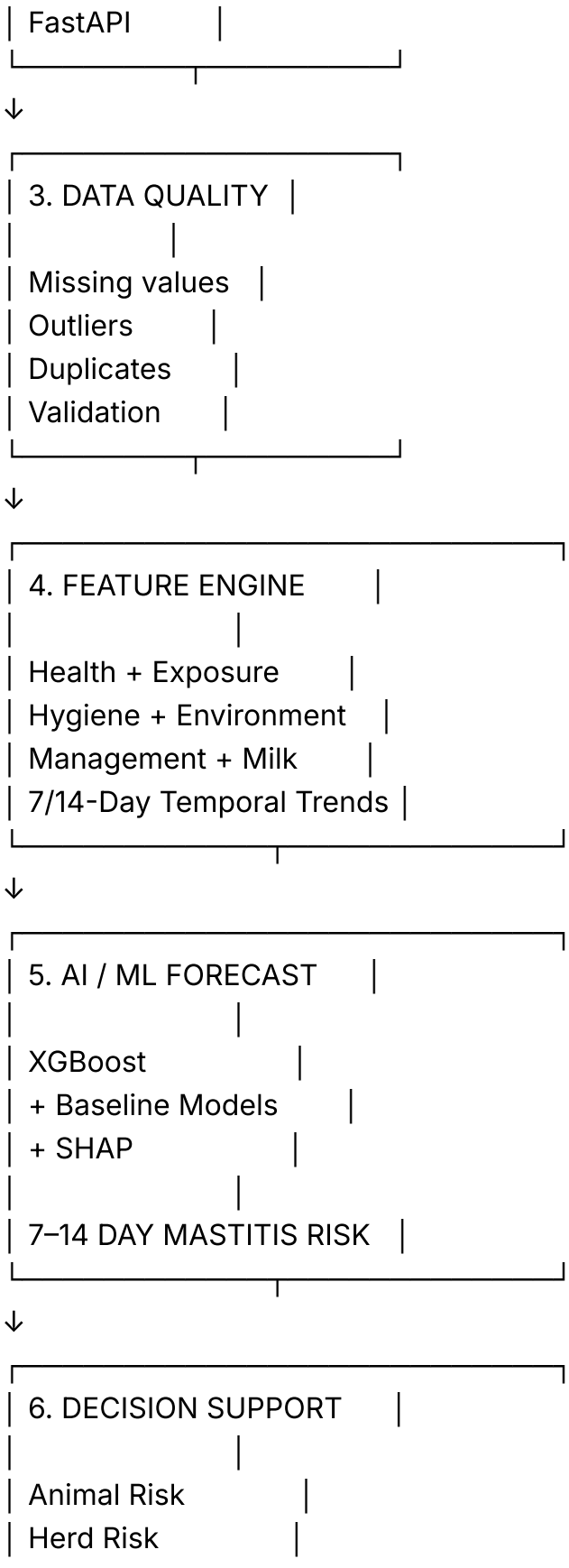

Cow ID: 1042
Risk: HIGH
Forecast: 7–14 days
Main factors:
• Wet bedding
• Hygiene exposure
• Previous history
• Milk trend
Shed A
Risk concentration
High-risk animals
Exposure hotspots
8. EARLY WARNING SYSTEM
HIGH RISK
↓
Farmer Mobile App
Veterinarian Dashboard
Farm Dashboard
SMS / Push Notification
9. PREVENTIVE ACTION ENGINE
Environmental Risk → Check Bedding
Hygiene Risk → Review Disinfection
Equipment Risk → Inspect Equipment
Animal Risk → Screen Animal
AI provides decision support;
veterinary confirmation remains needed.
10. CONFIRMATION
CMT / SCC / LAB / VETERINARIAN
CONFIRMED OUTCOME
11. MODEL EVALUATION & FEEDBACK

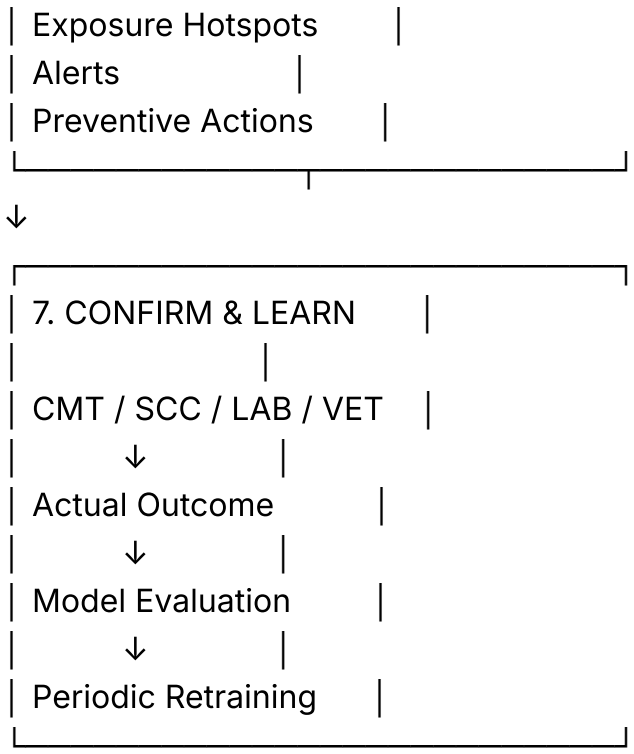
```
│  Predicted vs Actual                        │
│  • Precision                                │
│  • Recall / Sensitivity                     │
│  • Specificity                              │
│  • F1-score                                 │
│  • PR-AUC / ROC-AUC                         │
│  • 7–14 day Lead Time                       │
│  • False Alarm Rate                         │
│                                             │
│              ↓                              │
│      Confirmed Farm Data                    │
│              ↓                              │
│      Periodic Retraining                    │
│              ↓                              │
│        Validated Model                      │
└─────────────────────┬───────────────────────┘
                      │
                      └──────────────→ ML ENGINE
```

PPT VERSION — 7 MAJOR BLOCKS

For your actual architecture slide, don't put the giant 11-block diagram above. Use this:

```
┌──────────────────────────┐
│ 1. DATA                  │
│    COLLECTION            │
│                          │
│ RFID                     │
│ Hygiene                  │
│ Environment              │
│ Milk / Lab               │
│ Health / Farm            │
└──────────┬───────────────┘
           ↓
┌──────────────────────────┐
│ 2. EDGE &                │
│    INTEGRATION           │
│                          │
│ ESP32                    │
│ Raspberry Pi             │
│ MQTT / Wi-Fi             │
```


FastAPI
3. DATA QUALITY
Missing values
Outliers
Duplicates
Validation
4. FEATURE ENGINE
Health + Exposure
Hygiene + Environment
Management + Milk
7/14-Day Temporal Trends
5. AI / ML FORECAST
XGBoost
+ Baseline Models
+ SHAP
7–14 DAY MASTITIS RISK
6. DECISION SUPPORT
Animal Risk
Herd Risk



The one-line story for the slide

“Multi-source farm data → exposure & health features → 7/14-day temporal ML → individual & herd risk → early warning → preventive action → veterinary confirmation → model improvement.”

This is the version I’d use as your final architecture. It keeps your exposure/transmission idea, but removes the unnecessary complexity and avoids claiming that any individual sensor or the exposure graph directly diagnoses or proves transmission.